Learner-generated examples

Testing concept boundaries

Use learner-generated examples to test the boundaries of a concept. Constraints help students move beyond the first obvious example and expose what they think counts.

How to do it well

1 Choose the concept. Identify the defining feature students should apply and confirm they have encountered it.

2 Generate a first example. Give silent time and collect a bounded response before sharing suggestions.

3 Broaden the range. Ask for another valid example, then an unusual example or one satisfying an added constraint.

4 Test the boundary. Ask for a convincing non-example and an explanation of the feature that excludes it.

5 Compare and refine. Discuss selected cases, correct the definition where necessary and test a fresh constraint individually.

What this sounds like

“Give one example, another example and one that no one else is likely to choose. Finally, give a non-example that could fool someone, and explain the difference.”

Classroom example. In Languages, students generate a valid sentence, a second valid example and one satisfying an added constraint. They then write a plausible incorrect example and explain the rule it breaks. The range reveals whether pupils understand the grammatical boundary or rely only on the first familiar sentence pattern.

Top tips that make the technique work

Ask for several examples. A second or unusual example reveals more than a single familiar case.

Add purposeful constraints. Specify a feature that forces students to reconsider the concept rather than repeat their first answer.

Include a plausible non-example. Ask for a case someone might think valid and explain why it is not.

Inspect boundary reasoning. Students should justify membership using the definition rather than how unusual the example looks.

Teach the concept first. Generating examples is most useful when students have enough knowledge to test the constraints.

Do and avoid

Do	Avoid
Use constraints that expose the defining feature.	Unbounded example generation with no criterion.
Require explanations as well as examples.	Treating unusual answers as automatically sophisticated.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton — Learner-Generated Examples